ASSIGNMENTS

Problem set 1

AI504 — Knowledge Representation

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Contents

Problem 1

An *interval* is an ordered pair of real numbers (a, b) such that $a < b$. We say that the interval *interleaves* the interval (c, d) in case $a < c < b < d$. Is the "interleaves" relation transitive? If not, what is it transitive closure?

Solution

The relation definition is that

$$\text{if } (x, y) \in R \text{ then } x\boldsymbol{R}y.$$

The interleaves relation is defined as:

$$(a, b)\boldsymbol{R}(c, d) = a < c < b < d.$$

For transitivity, it is required that

$$(a, b)\boldsymbol{R}(c, d) \text{ and } (c, d)\boldsymbol{R}(e, f) \Rightarrow (a, b)\boldsymbol{R}(e, f).$$

By the interleaves definition, we know that

$$(a, b)\boldsymbol{R}(c, d) = a < c < b < d$$

$$(c, d)\boldsymbol{R}(e, f) = c < e < d < f$$

we can call these our premises.

Now we wish to prove that

$$(a, b)\boldsymbol{R}(e, f) = a < e < b < f.$$

For this, there are the following requirements

$$a < c < e < b < d < f$$

then by our premises we can prove the following

$$a < c \text{ and } c < e \Rightarrow a < e$$

and

$$b < d \text{ and } d < f \Rightarrow b < f.$$

Since we only know that

$$c < e < d \text{ and } c < b < d$$

we cannot say anything about whether $e < b$ is true.

Therefore, the relation is **not transitive**.

Let's do a **counterexample.**

Let

$$A = [1, 3] \quad B = [2, 6] \quad C = [5, 8].$$

If we apply

$$A\boldsymbol{R}B \text{ and } B\boldsymbol{R}C \Rightarrow A\boldsymbol{R}C.$$

This would imply that $[1, 3]$ interleaves $[5, 8]$.

This is false because it would imply that

$$1 < \mathbf{5} < \mathbf{3} < 8.$$

For this to work we need the relation closure R^+.

Let's argue that for the closure, we need

$$(a, b)\boldsymbol{R}(c, d) \text{ and } (c, d)\boldsymbol{R}(e, f) \Rightarrow (a, b)\boldsymbol{R}(e, f)$$

where

$$(a, b)\boldsymbol{R}(e, f) = a < e < b < f$$

Since by definition we already know $a < e$ and $b < f$.

The closure needs to be

$$R^+ = \{(e, b)\}.$$

Problem 2

Let x and Y be sets, let R and S be binary relations of type $X \times Y$, and let P and Q be subsets of X. Recall that the *forward image* $R(P)$ is the subset of Y obtained by following all R-arrows forwards out of P. (And similarly for all the other forward images in this problem.) For each of the following identities, either prove correct or provide a counterexample. *All proofs are short and all counterexamples are tiny*

(a) $R(P \cup Q) = R(P) \cup R(Q)$

(b) $R(P \cap Q) = R(P) \cap R(Q)$

(c) $(R \cup S)(P) = R(P) \cup S(P)$

(d) $(R \cap S)(P) = R(P) \cap S(P)$

Solution

A relation is a relationship between sets of values such that for a set X

$$R \subseteq \{(x, y) \mid x, y \in X\}$$

(a)

We wish to show that

$$R(P \cup Q) = R(P) \cup R(Q)$$

We know that

$$\forall y \in R(P \cup Q), \exists x \in P \cup Q \mid (x, y) \in R$$

This means that for every y in the relation on the union of the subsets P and Q, there must be some x where the relation R holds.

Then we can show that the y can either come from $R(P)$ or $R(Q)$, so

$$\forall y \in R(P) \cup R(Q) : y \in R(P) \text{ or } y \in R(Q).$$

This is the same as $R(P \cup Q) = R(P) \cup R(Q)$.

$\square$

(b)
We wish to show that

$$R(P \cap Q) = R(P) \cap R(Q).$$

The same applies for

$$\forall y \in R(P \cap Q), \exists x \in P \cap Q \mid (x, y) \in R$$

again the y must come from an x mapped R.

But when we do

$$\forall y \in R(P) \cap R(Q) : y \in R(P) \text{ and } y \in R(Q).$$

We realise that for y to exist, P and Q must share an element that both maps to the same y, if they don't then **this does not hold**.

Let's do a **counterexample** to properly disprove this.

Let's check if

$$R(P \cap Q) = R(P) \cap R(Q)$$

when

$$X = \{1, 2\} \text{ and } Y = \{1, 2\} \text{ and } P = \{1\} \subseteq X \text{ and } Q = \{2\} \subseteq X.$$

Now let

$$R = \{(1, 2), (2, 2)\}$$

then

$$R(P \cap Q) = \{(1, 2), (2, 2)\}(\{1\} \cap \{2\}) = \emptyset$$

and

$$R(P) \cap R(Q) = \{(1, 2), (2, 2)\}(\{1\}) \cap \{(1, 2), (2, 2)\}(\{2\}) = \{2\} \cap \{2\} = \{2\}.$$

Since $\emptyset \neq \{2\}$ this **counters** the statement.

$\square$

(c)
We wish to show that

$$(R \cup S)(P) = R(P) \cup S(P)$$

we can say that any y must come from an x in P applied by R or S

$$\forall y \in (R \cup S)(P), \exists x \in P \mid (x, y) \in R \text{ or } S.$$

Then we can describe $y \in R(P) \cup S(P)$

$$\forall y \in R(P) \cup S(P), \exists x \in P| \ (x, y) \in R \text{ or } S.$$

Since these both describe the same relation from $x \to y$ they are **equal**.

$\square$

(d)
We wish to show that

$$(R \cap S)(P) = R(P) \cap S(P)$$

any y must have an x from P which is applied by $R \cap S$

$$\forall y \in (R \cap S)(P), \exists x \in P \mid (x, y) \in R \cap S.$$

Then showing that x can only be

$$\forall y \in R(P) \cap S(P), \exists x \in P \ |(x, y) \in R \text{ and } S.$$

Much like in (b) we realise that for y to exist, R and S must share an element that both maps the same x to the same y, if they don't then **this does not hold**.

Let's do a **counterexample** to disprove this.

Let's check if

$$(R \cap S)(P) = R(P)) \cap S(P)$$

$$X = \{1, 2, 3\} \text{ and } Y = \{1, 2, 3\} \text{ and } P = \{1, 2\} \subseteq X.$$

Now let

$$R = \{(1, 2), (1, 3), (3, 1)\} \quad S = \{(1, 3), (2, 1), (2, 2), (3, 3)\}$$

then

$$R \cap S = \{(1, 3)\}.$$

Now we can check

$$(R \cap S)(P) = \{(1, 3)\}(\{1, 2\}) = \{3\}$$

and

$$\begin{aligned} R(P)) \cap S(P) &= \{(1, 2), (1, 3), (3, 1)\}(\{1, 2\})) \cap \{(1, 3), (2, 1), (2, 2), (3, 3)\}(\{1, 2\}) \\ &= \{2, 3\} \cap \{1, 2, 3\} = \{2, 3\}. \end{aligned}$$

Since $\{3\} \neq \{2, 3\}$ this **counters** the statement. $\square$

Problem 3

A *full binary* tree is a nonempty binary tree where each node is a leaf or has exactly two children. Define a function $\jmath$ from full binary trees to rational numbers by recursion as follows:

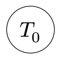
$$\jmath(T) := \begin{cases} 1 & \text{if } T \text{ is a leaf} \\ 1 + \frac{\jmath(T_0)+\jmath(T_1)}{4} & \text{if otherwise, where } T_0 \text{ and } T_1 \text{ are the two subtrees} \end{cases}$$

Calculate the value of $\jmath$ on lots of small examples. Formulate a conjecture about how big it can get, and prove it.

Solution

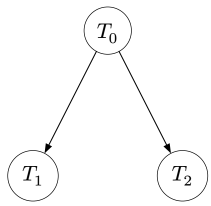
Let's start off by doing a few examples

Example 1



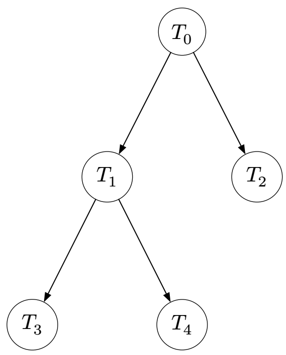
Here $\jmath(T) = 1$

Example 2



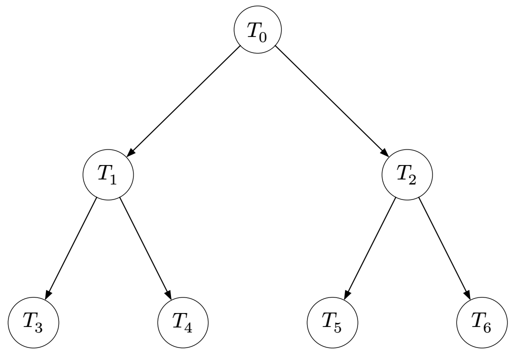
Here $\jmath(T) = 1 + \frac{1+1}{4} = 1 + \frac{1}{2} = 1.5$

Example 3



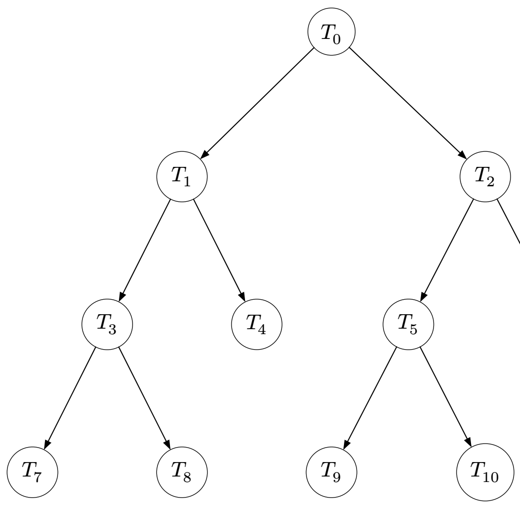
Here $\jmath(T) = 1 + \frac{(1+\frac{1+1}{4})+1}{4} = 1 + \frac{2+\frac{1}{2}}{4} = 1.625$

Example 4



Here $\jmath(T) = 1 + \frac{(1+\frac{1+1}{4})+(1+\frac{1+1}{4})}{4} = 1 + \frac{3}{4} = 1.75$

Example 5



Here $\jmath(T) = 1 + \frac{\left(1+\frac{\left(1+\frac{1+1}{4}\right)+1}{4}\right)+\left(1+\frac{\left(1+\frac{1+1}{4}\right)+1}{4}\right)}{4} = 1 + \frac{13}{16} = 1.8125$

Conjecture

From the examples above, it seems that $\jmath(T_{\text{size } k}) \to 2$ as $k \to \infty$.

Therefore, for any finite tree T, it seems that $\jmath$ grows infinitely close to 2. Therefore:

$$\jmath(T) < 2$$

Let's try and prove that

Proof by structural induction

We want to prove that $\jmath(T) < 2$

<u>**Base case**</u>

if T_0 is a single leaf (a node with no children) then $\jmath(T) = 1$

$$\jmath(T_0) < 2.$$

<u>**Inductive hypothesis**</u>

Assume that

$$\jmath(T_{\text{subtree 1}}) < 2 \text{ and } \jmath(T_{\text{subtree 2}}) < 2$$

<u>**Inductive step**</u>

For any tree with two subtrees (full tree)

$$\jmath(T) = 1 + \frac{\jmath(T_{\text{subtree 1}}) + \jmath(T_{\text{subtree 2}})}{4}.$$

Since, by ***IH***

$$\jmath(T_{\text{subtree 1}}) + \jmath(T_{\text{subtree 2}}) < 4.$$

A new tree, composed of two subtrees, becomes

$$\jmath(T) = 1 + \frac{\jmath(T_{\text{subtree 1}}) + \jmath(T_{\text{subtree 2}})}{4}.$$

Since $\frac{\jmath(T_{\text{subtree 1}}) + \jmath(T_{\text{subtree 2}})}{4} < 1$

$$\jmath(T) = 1 + \frac{\jmath(T_{\text{subtree 1}}) + \jmath(T_{\text{subtree 2}})}{4} < 2$$

Problem 4

Consider a directed graph G with 12 vertices that satisfies the following two properties:

- G is *strongly connected*, i.e., between any two distinct vertices u and v, there is a directed path from u to v as well as a directed path from v to u, and
- there is at most one directed edge coming out of each vertex.

Draw G, and argue why your drawing is the *only possible* solution (up to isomorphism).

Solution

Since each vertex has **at most one** directed edge going out, and G must be strongly connected, every vertex must have **exactly one** outgoing as well as **exactly one** ingoing edge as every node must also be reachable. This creates a 12-cycle:

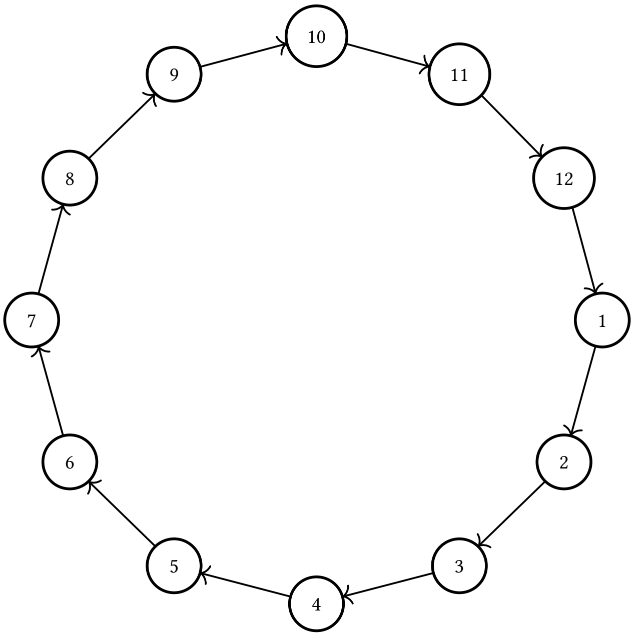


Figure 1: G : A strongly connected 12-cycle.

To say that Figure 1 is the *only possible solution* (up to isomorphism). We need to argue that Figure 1 is isomorphic.

By definition, two directed graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are **isomorphic** if and only if there exists a one-to-one and onto function $f : V_1 \to V_2$ such that any two vertices $(a, b) \in V_1$ where $(a, b) \in E_1$ and $(f(a), f(b)) \in V_2$ where $(f(a), f(b)) \in E_2$. [1]

For a function to be one-to-one and onto on two sets, it is at least required that they have the same amount of elements. For this 12-cycle, any *other solution* would, by definition, have exactly 12 vertices as well.

For this 12-cycle, because its vertices have exactly one ingoing and exactly one outgoing edge, the adjacency property of isomorphism **can** be satisfied. Any two vertices in two in a strongly connected cycle like this, will hold this property. The only way this protery would **not** hold, is that a vertex in the graph has either 2 outgoing edges or 2 ingoing edges.

This can be showed with an example of 2 isomorphic graphs:

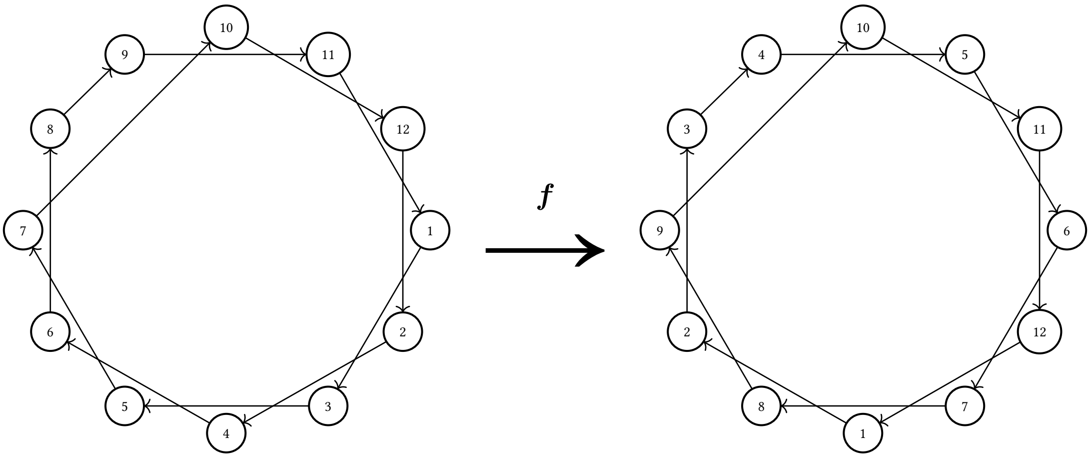


Figure 2: G : Two isomorphic directed graphs due to a function f

Here, the left graph fullfills the same requirements as directed graph G on Figure 1, though it has different relations. By applying the following one-to-one and onto function f to the graph:

$$f(12) = 11, f(2) = 12, f(4) = 1, f(6) = 2, f(8) = 3, f(9) = 4,$$
$$f(11) = 5, f(1) = 6, f(3) = 7, f(5) = 8, f(7) = 9, f(10) = 10$$

yields the same relations as the original graph G on Figure 1.

Because of this, Figure 1 depicts the ***only possible*** solution (up to isomorphism).

Bibliography

[1] K. H. Rosen, *Discrete Mathematics and Its Applications*, 8th ed. New York, USA: McGraw-hill education, 2019.